

① 3.5-2

$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 0 & -1 & -1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

$$\begin{bmatrix} \textcircled{1} & 1 & 1 & 0 & 0 & 0 \\ 0 & \textcircled{1} & 1 & 1 & 1 & 0 \\ 0 & 0 & \textcircled{1} & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$v_1, v_2 \text{ \& } v_3$

③

② $3.5 - 20$

$$x - 2y + 3z = 0$$

a) $\begin{bmatrix} (1) & -2 & 3 \end{bmatrix} x = 0$

$$N(A) = c_1 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Basis.

b) $N(A) \cap \text{row plane} \rightarrow$

$$\begin{bmatrix} (1) & -2 & 3 \\ 0 & 0 & (1) \end{bmatrix}$$

$$N(A) = \lambda \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

c) Basis for all $\perp \rightarrow \text{row space} \rightarrow \lambda \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$

③ 3.5 - 37

$$AS = SA \quad \text{shift matrix}$$

$$S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$AS = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & a & b \\ 0 & d & e \\ 0 & g & h \end{bmatrix} = \begin{bmatrix} d & e & f \\ g & h & i \\ 0 & 0 & 0 \end{bmatrix}$$

$$SA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} d & e & f \\ g & h & i \\ 0 & 0 & 0 \end{bmatrix}$$

$$d = 0 \quad g = 0 \quad h = 0$$

$$a = e \quad b = f \quad i = e$$

3 dimensions

Q) 3.5 - 41

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P \rightarrow c_1 \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$+ c_3 \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} + c_4 \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

$$+ c_5 \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} c_2 & c_1 + c_5 & c_3 + c_4 \\ c_1 + c_4 & c_3 & c_2 + c_5 \\ c_3 + c_5 & c_2 + c_4 & c_1 \end{bmatrix}$$

$$\begin{bmatrix} c_2 & c_1 + c_5 & c_3 + c_4 \\ c_1 + c_4 & c_3 & c_2 + c_5 \\ c_3 + c_5 & c_2 + c_4 & c_1 \end{bmatrix}$$

- $c_1 = 1$ $c_2 = 1$ $c_3 = 1$
 $c_4 = -1$ $c_5 = -1$ for I

- For zero

$$\begin{array}{lll} c_1 = 0 & c_2 = 0 & c_3 = 0 \\ c_4 = 0 & c_5 = 0 & \end{array}$$

Only solution $c_1 = c_2 = \dots = 0$

⑤ 3.5-44

$$\dim(V) + \dim(W) = \dim(V \cap W) + \dim(V + W)$$

$$v \in V$$

$$w \in W$$

$$[v \quad w] \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \boxed{c_1 v + c_2 w}$$

$\begin{bmatrix} c_{11} \\ c_{12} \\ \vdots \end{bmatrix}$
 $\begin{bmatrix} c_{21} \\ c_{22} \end{bmatrix} \quad V + W$

$$A \rightarrow m \times n$$

$$\text{rank } r$$

$$\begin{aligned} \dim C(A) &= r \\ \dim N(A) &= n - r \\ \dim(\text{input}) &= n \end{aligned} \quad + \quad \underline{\underline{\quad}}$$

Q 3.5-43

$$V \cap W = u_1, \dots, u_r$$

$$V = u_1, \dots, u_r, v_1, \dots, v_s$$

$$W = u_1, \dots, u_r, w_1, \dots, w_t$$

$$\dim(V \cap W) = r$$

$$\dim(V) = r + s$$

$$\dim(W) = r + t$$

$$\begin{aligned} \dim(V + W) &= r + s + \cancel{r} + t - \cancel{r} \\ &= r + s + t \end{aligned}$$

$$\dim(V + W) = r + s + t$$

Only possible when u, v, w are combined to form a basis.

⑦ 3.6 - 11

$A \rightarrow m \times n$ rank r

$\Rightarrow$ There exist b for which no solution

$\hookrightarrow m > n$ $m > r$

a)

$$\begin{aligned} r &\leq n \\ n &\leq m \\ r &< m \end{aligned}$$

b) $A^T y = 0$

$$\dim C(A^T) = r$$

$$\dim N(A^T) = m - r$$

$$\underline{\dim N(A^T) \geq 1}$$

⑧ 3.6 - 24

$$A^T y = d$$

a) is solvable when d is in $C(A^T)$

b) Solution y is unique when nullspace of A^T contains only zero vector.

9) 2.6 - 28

$B \rightarrow \text{rank} \rightarrow 2$

$C \rightarrow \text{rank} \rightarrow 2$

a) <u>Basis for</u>	<u>Row space</u>	<u>Left nullspace</u>
B	$[1 \ 0 \ 1 \ 0 \ \dots]$	
	$[0 \ 1 \ 0 \ 1 \ \dots]$	
$\dots$	$\dots$	$\dots$
C	$[r \ n \ b \ q \ \dots]$	
	$[p \ p \ p \ \dots]$	

b) B is a symmetric matrix so same nullspace & left nullspace

$$B \rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & - & - & - & - \\ \vdots & & \underbrace{\hspace{2cm}} & & & & & \\ & & \downarrow & & & & & \end{bmatrix}$$

6 free vars.

$$N(B) = c_1 \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ 1 \\ 0 \end{bmatrix} \dots$$

$$\text{rref}(C) \rightarrow \begin{bmatrix} r & n & b & q & k & b & n & r \\ r & r & r & r & r & r & r & r \\ & & & & & & & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} r & n & b & - & - & - \\ 0 & r-n & r-b & - & - & - \end{bmatrix}$$

$$\begin{bmatrix} r & n & b & q & k & b & n & r \\ 0 & (r-n) & r-b & r-q & r-k & r-b & r-n & 0 \end{bmatrix}$$

$$N(c) = c_1 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} \frac{n-b}{r-n} \\ \frac{b-k}{r-n} \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

dimension $\rightarrow 6$

⑩ 2.6 - 30

$$A = u v^T$$

$$A > 2 \times 2$$

$$\text{rank} = 1$$

$$u \rightarrow \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \quad v \rightarrow \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$u v^T = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} u_1 v_1 & u_1 v_2 \\ u_2 v_1 & u_2 v_2 \end{bmatrix}$$

$$\rightarrow u v^T x = 0$$

$$v^T x = 0$$

$$u_2 v_2 + u_2 v_1 = \lambda (u_1 v_1 + u_1 v_2)$$

$$u_2 (\cancel{v_1} + v_2) = \lambda u_1 (\cancel{v_1} + v_2)$$

$$\boxed{u_1 = c u_2}$$

$$\rightarrow \begin{bmatrix} \cdot & v_2/v_1 \\ 0 & 0 \end{bmatrix}$$

$$\text{colspace} \rightarrow \lambda \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$N(A) = \lambda \begin{bmatrix} -v_2 \\ v_1 \end{bmatrix}$$

$$\text{row space} \rightarrow \lambda v$$

⑪ 3.6 - 3.1

$M \rightarrow$ space of 3×3 matrices

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \quad A \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$$

a) x for which $Ax = 0$

$$x \rightarrow c_1 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad c_2 \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

dim $\rightarrow 3$

b) $Ax \rightarrow 3 \times 3$ matrix

$$Ax = B$$

$$x = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} a-g & b-h & c-i \\ d-a & e-b & f-c \\ g-d & h-e & i-f \end{bmatrix}$$

$$\rightarrow (Ax) \rightarrow E(Ax)$$

$$\begin{bmatrix} a-g & b-h & c-i \\ d-a & e-b & f-c \\ 0 & 0 & 0 \end{bmatrix} \quad 6 \text{ independent components}$$